Consequences of Gravel Mining: Tracing the Chemical Footprint in a Local Watershed

Anna Sophia Callens[1], Gal Koss[2]
[1] Pioneer High School
[2] University of California - Davis, Department of Plant Sciences

Introduction

Mining throughout California became industrialized during the Gold Rush era, and today it is critical for advancing construction efforts. The recent extension of CEMEX's gravel mining permit to continue operations in Cache Creek threatens Yolo County's local watershed health as it creates dangerous water quality conditions for aquatic life and human health.

Sulfuric acid comes from mining exposing sulfur bearing materials to oxygen, water, and iron-oxidizing bacteria which makes an acidic solution with high heavy metals. Therefore, mining lowers the water's acidity, and an unstable pH can be harmful to aquatic organisms.

The acidic water byproduct comes from the sulfuric acid that dissolve heavy materials, and this dissolving or decomposition needs oxygen in the reaction. This is what leads to lower oxygen levels in the water, causing the dissolved oxygen to be decreased due to these mining operations[1].

Another key byproduct of mining is mercury. Mercury contamination is an existing issue in Northern California waterways, and continuing the introduction of mercury into local creeks will proceed to jeopardize aquatic ecosystems.

Hypothesis:

Compared to a control site upstream of the Cache Creek gravel mine, downstream sites have lower dissolved oxygen and pH levels, and higher mercury levels.

Motivation

This image was taken along Cache Creek while taking my samples. It shows how mercury contamination is still a problem recognized by my community. However, the sign was placed underneath a bridge out of sight, demonstrating how this problem is not widely known amongst the public.

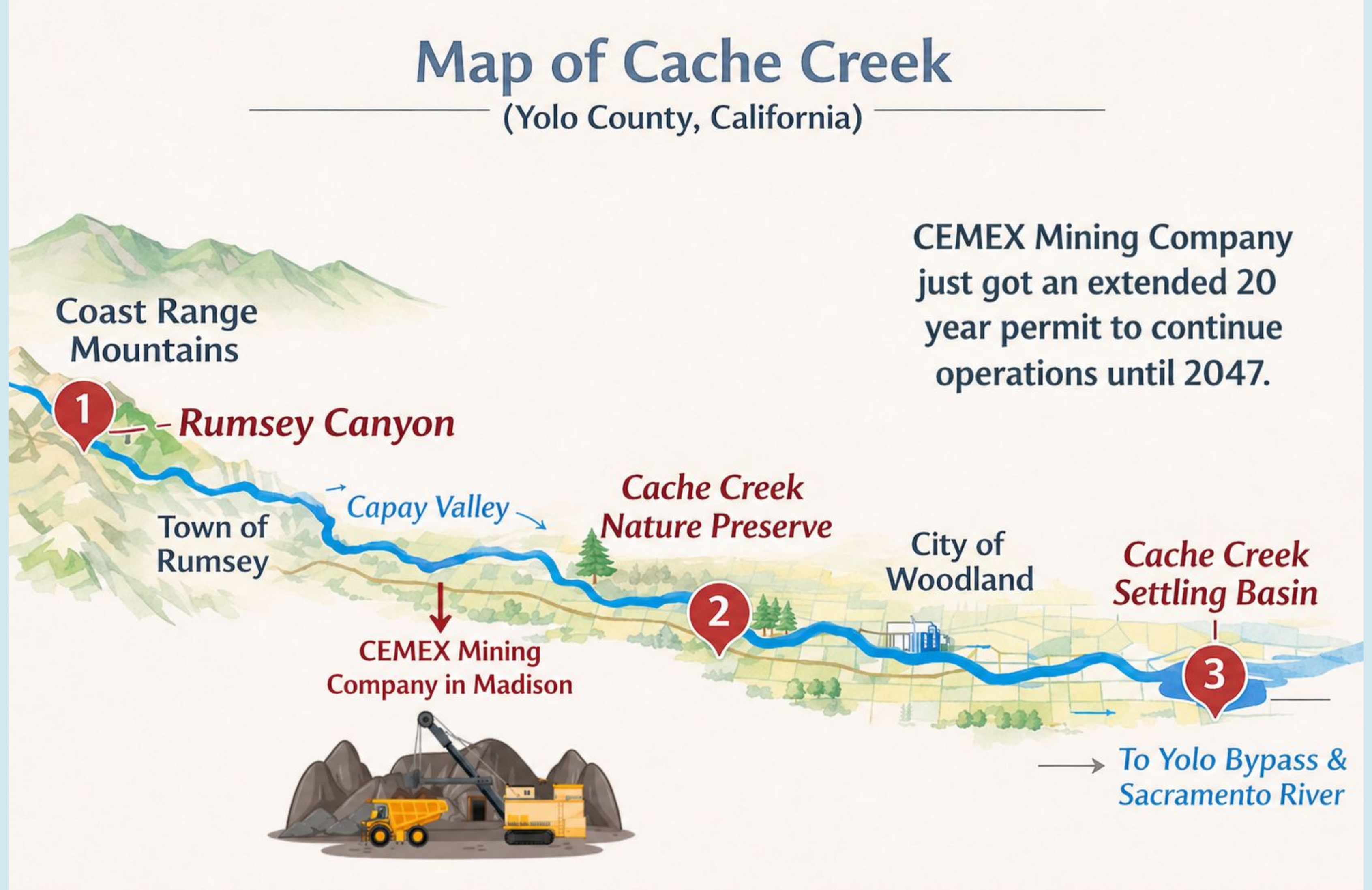
CEMEX Gravel Mining Company is located in multiple Northern California sites besides Yolo County such as Fresno County and Contra Costa County. A meeting was held to discuss the extension of gravel mining in Yolo County. Supervisors voted unanimously to extend CEMEX's permit[2].

UCDAVIS
UNIVERSITY OF CALIFORNIA

Global Council for Science and the Environment
EnvironMentors

NSF

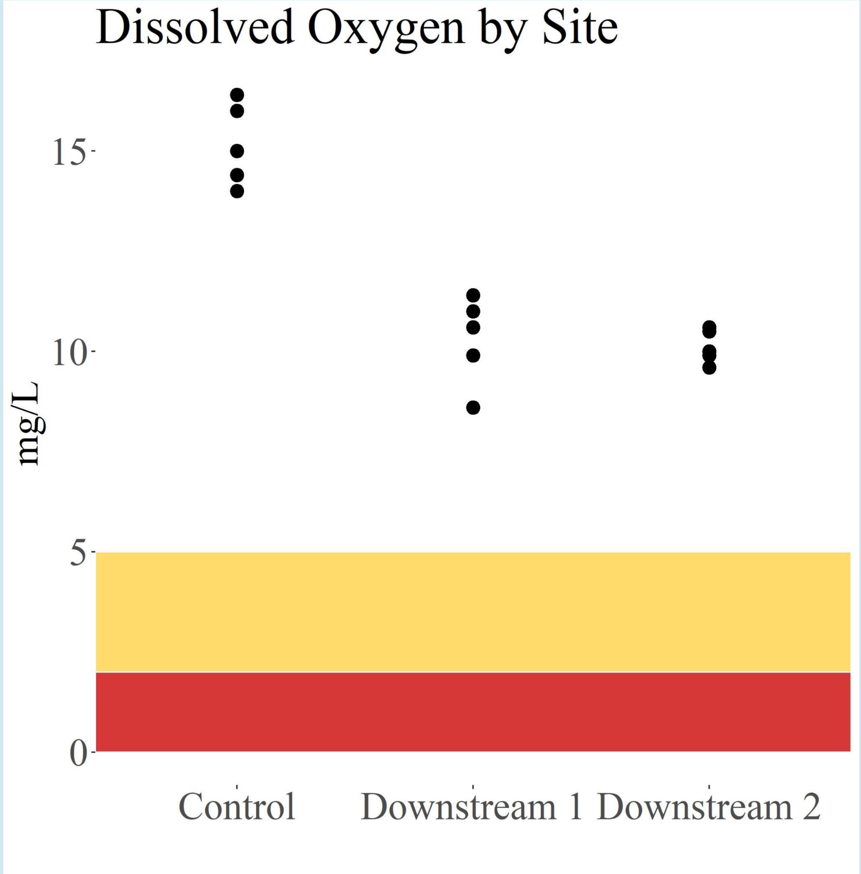
Methods



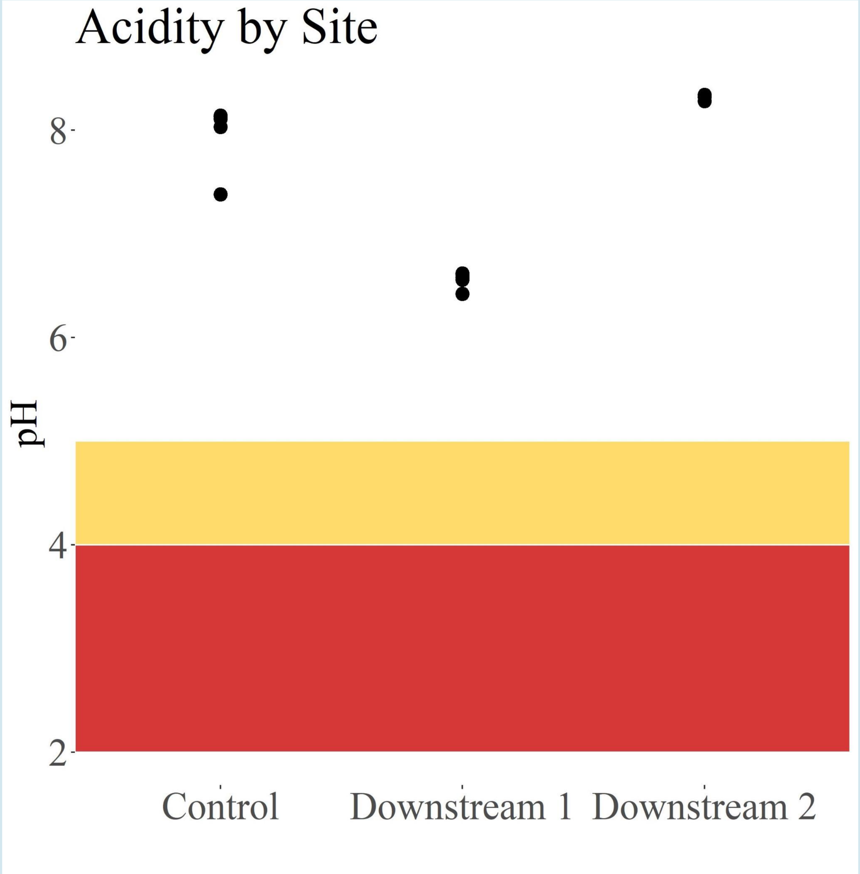
I collected data from 3 sites along Cache Creek with Rumsey Canyon being the control site, Cache Creek Nature Preserve being the immediate treatment site as it is located immediately downstream of the local CEMEX Gravel Mine, and Cache Creek Settling Basin as the secondary treatment site as it is further downstream from the mine but still affected by mining operations. The dissolved oxygen levels were tracked using the Milwaukee 600 Meter and the pH levels were found using the Mettler Toledo machine. To test mercury levels within our budget, we used an online DIY strip kit that had colors change based on the concentration that ranged from 0.0 to 0.01.

Results

Preliminary Results: The color of the mercury strip was in the safe range set by the USEPA (less than 0.002). However, we observed that the strip taken at downstream site 1 (initial treatment) had the darkest coloration, showing that it had the strongest presence of mercury between the three strips.



Gravel mining decreases dissolved oxygen in both downstream treatment sites but does not go below an immediate dangerous level.



Gravel mining lowers pH in the first treatment site but does not go below the immediate dangerous level.

Conclusion

- Low dissolved oxygen and low pH are both factors that stress out and harm aquatic life. Although the tests did not go beneath the immediate dangerous levels of pH and DO, it is concerning that mining operations affected it.
- High mercury levels can cause adverse health effects. Many studies have proved Cache Creek's mercury contamination, especially considering there is signage posted at certain spots along the creek. Using an at home DIY kit, I was not able to get an exact number which raises concerns as to whether the public could be able to pinpoint mercury levels themselves.

➔ Factors such as rain, different land settings, and spaced out sampling were all limitations to note that could have affected the outcome of this project.

Low dissolved oxygen, low pH, and high mercury levels create optimal conditions for the compound methylmercury (MeHg) to be produced. Methylmercury is one of the worst outcomes regarding the effects of mercury contamination. Most of these mining sources have a non-toxic inorganic form of mercury as a byproduct, however, when it reaches a wetted environment like a reservoir, it settles to the bottom where bacteria in the sediments convert it to the more toxic organic form that is methylmercury[3].

Since it is a bioaccumulative pollutant, it increases in concentration as it goes up through the food chain. It can go from algae to zooplankton to prey fish and then to their predators. This not just an alarming issue for the creek's organisms, it is also a problem for human health as a major exposure pathway for humans to mercury is fish consumption.

- "80% to 90% of organic mercury in a human body is from fish and shellfish intake, and 75% to 90% of organic mercury existing in fish and shellfish is methylmercury[4]."

Why choose Cache Creek?

The USGS and DWR have been monitoring water quality at the Cache Creek Settling Basin since 2010, after it was discovered that Cache Creek is a significant contributor of mercury and methylmercury to the Sacramento-San Joaquin Delta through the Yolo Bypass. The California Regional Water Quality Control Board has estimated that the Cache Creek supplies 10-30% of the mercury loads into the Sacramento River system, though it represents only about 2% of the water flow[5]."

This is an immediate problem, especially for indigenous tribes that populate Northern California. Studies have shown that MeHg exposure of tribal populations from fish are about 3 to 10 times higher than the US general population and that exposure poses potential health risks. The California Indigenous Environmental Alliance was created in part to address the ongoing problem of mercury exposure in these tribal communities as their surveys indicate that tribes have reduced rates of fish consumption because of fear of mercury contamination[6].

Since its founding, the C.I.E.A. has worked to educate communities about the health risks of mercury. However, education is not enough to resolve this problem. Stricter policies must be put in place for gravel mining companies to address the issue.

References:

[1]Nancucheo, I., Bitencourt, J. A. P., Sahoo, P. K., Alves, J. O., Siqueira, J. O., & Oliveira, G. (2017). Recent Developments for Remediating Acidic Mine Waters Using Sulfidogenic Bacteria. BioMed Research International, 2017, 1–17. https://doi.org/10.1155/2017/7256582
[2]Hennessy, D. (2025, December 10). Yolo County extends controversial gravel mining permit along Cache Creek. Abridged – PBS KVIE. https://www.abridged.org/news/yolo-county-extends-gravel-mining-permit-along-cache-creek/
[3](2024, September 2). EXPLAINER: Mercury contamination in California's waterways. MAVEN'S NOTEBOOK | California Water News Central. https://mavensnotebook.com/explainers/mercury-contamination-in-californias-waterways/
[4]Hong, Y.-S., Kim, Y.-M., & Lee, K.-E. (2012). Methylmercury Exposure and Health Effects. Journal of Preventive Medicine & Public Health, 45(6), 353–363. https://doi.org/10.3961/jpmph.2012.45.6.353
[5]Mercury's Lasting Legacy: Measuring Contamination in Cache Creek. (2017). USGS. https://www.usgs.gov/news/mercurys-lasting-legacy-measuring-contamination-cache-creek
[6]Xue, J., Zartarian, V., Mintz, B., Weber, M., Bailey, K., & Geller, A. (2015). Modeling tribal exposures to methyl mercury from fish consumption. Science of the Total Environment, 533, 102–109. https://doi.org/10.1016/j.scitotenv.2015.06.070